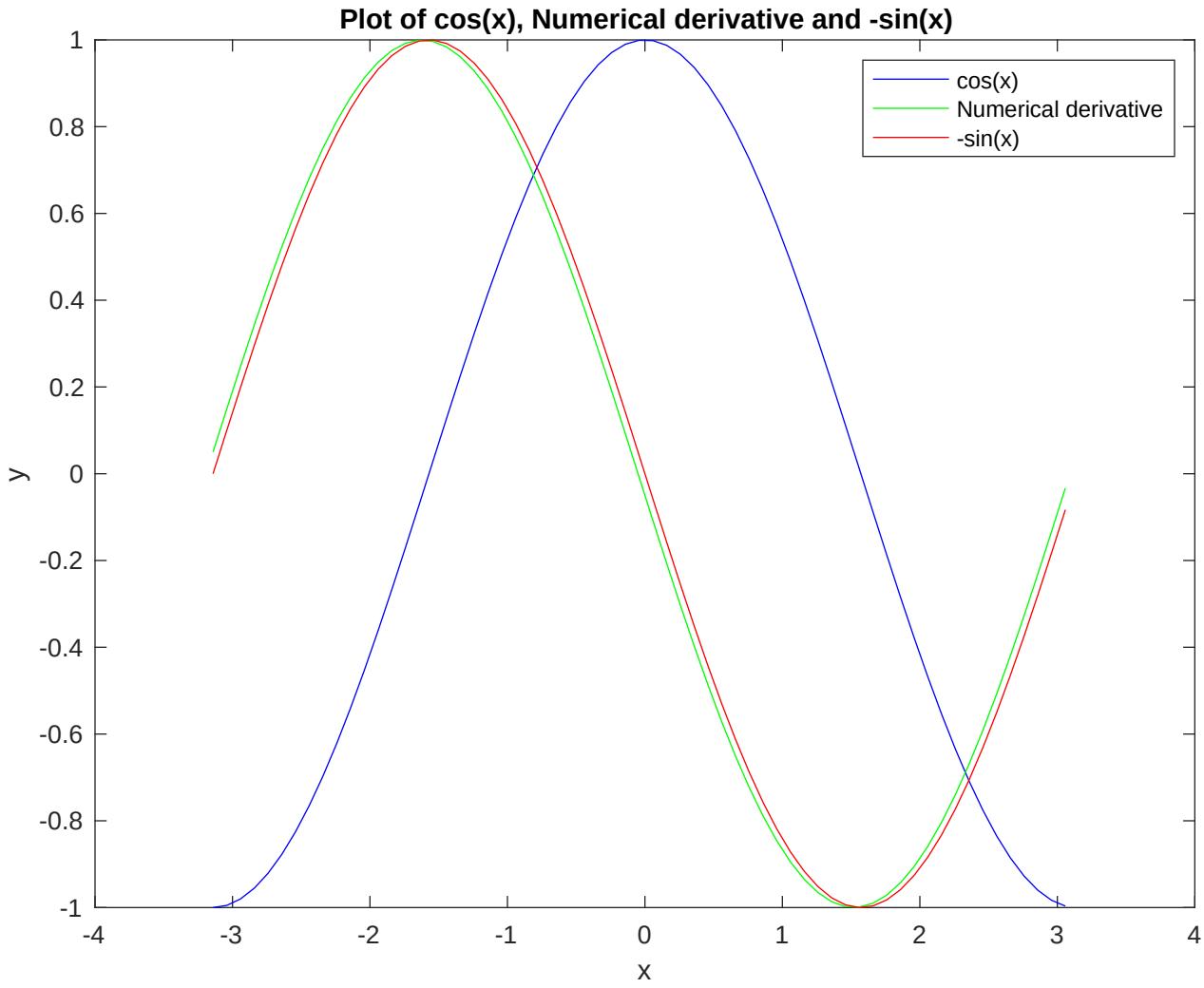
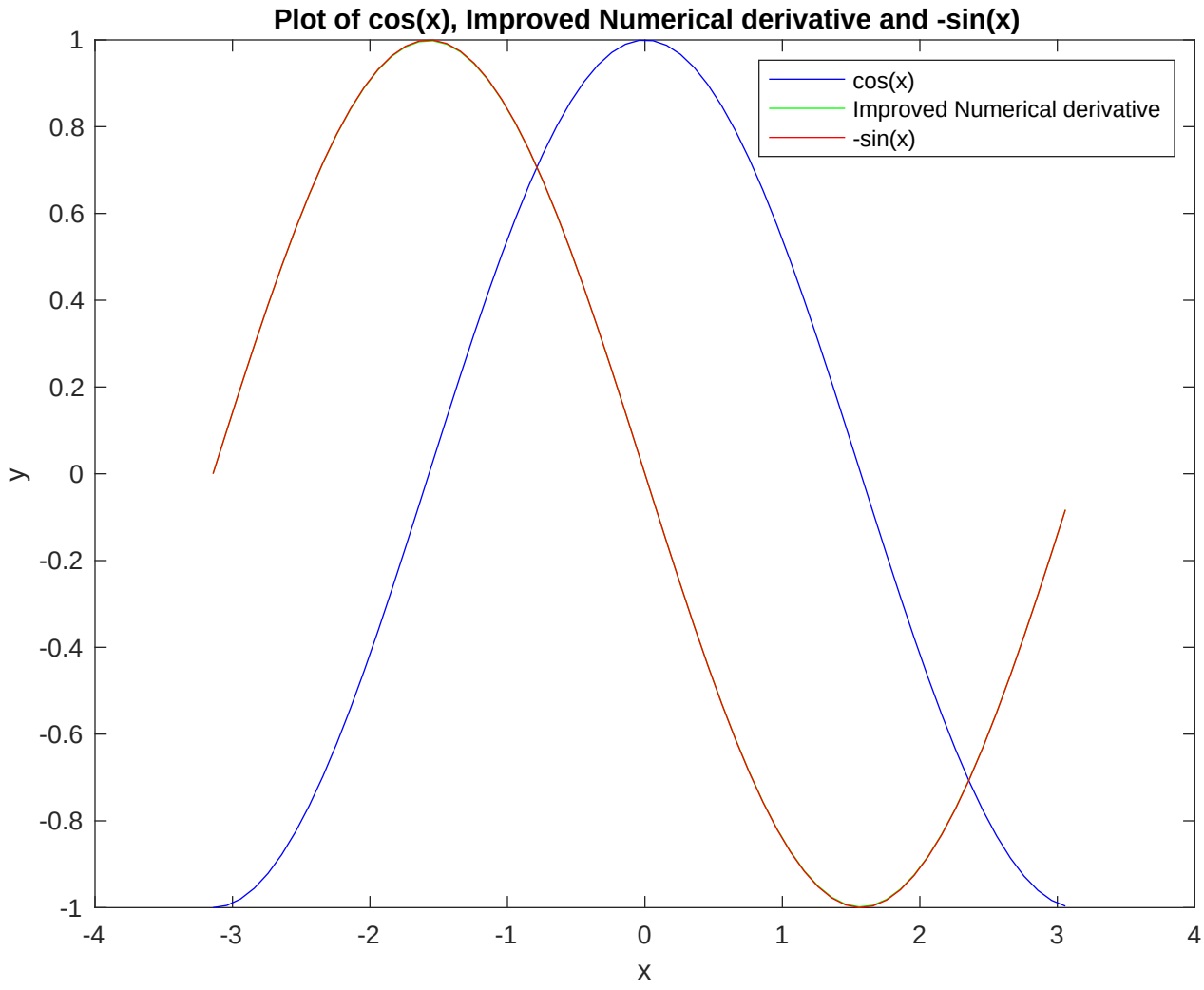
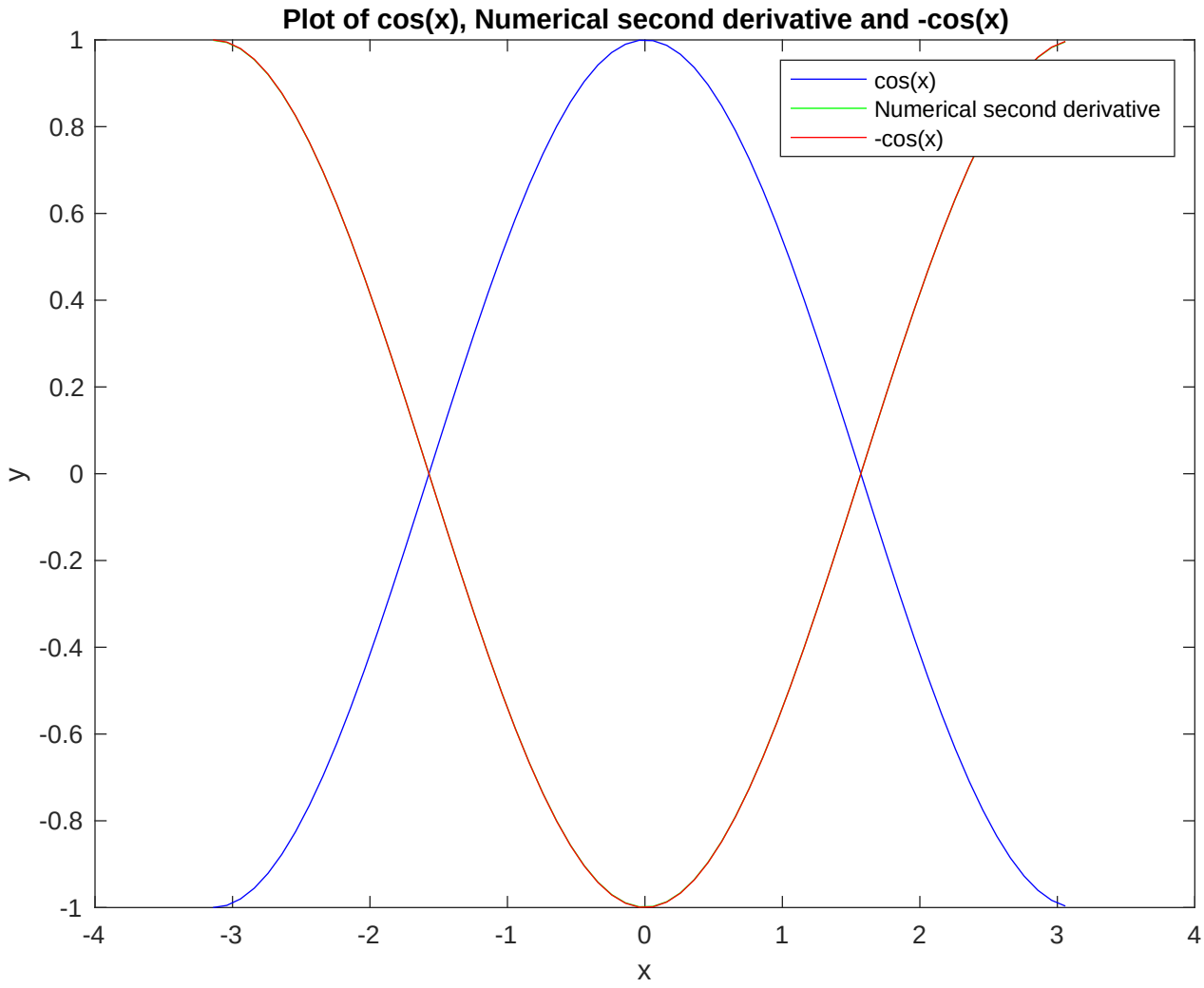
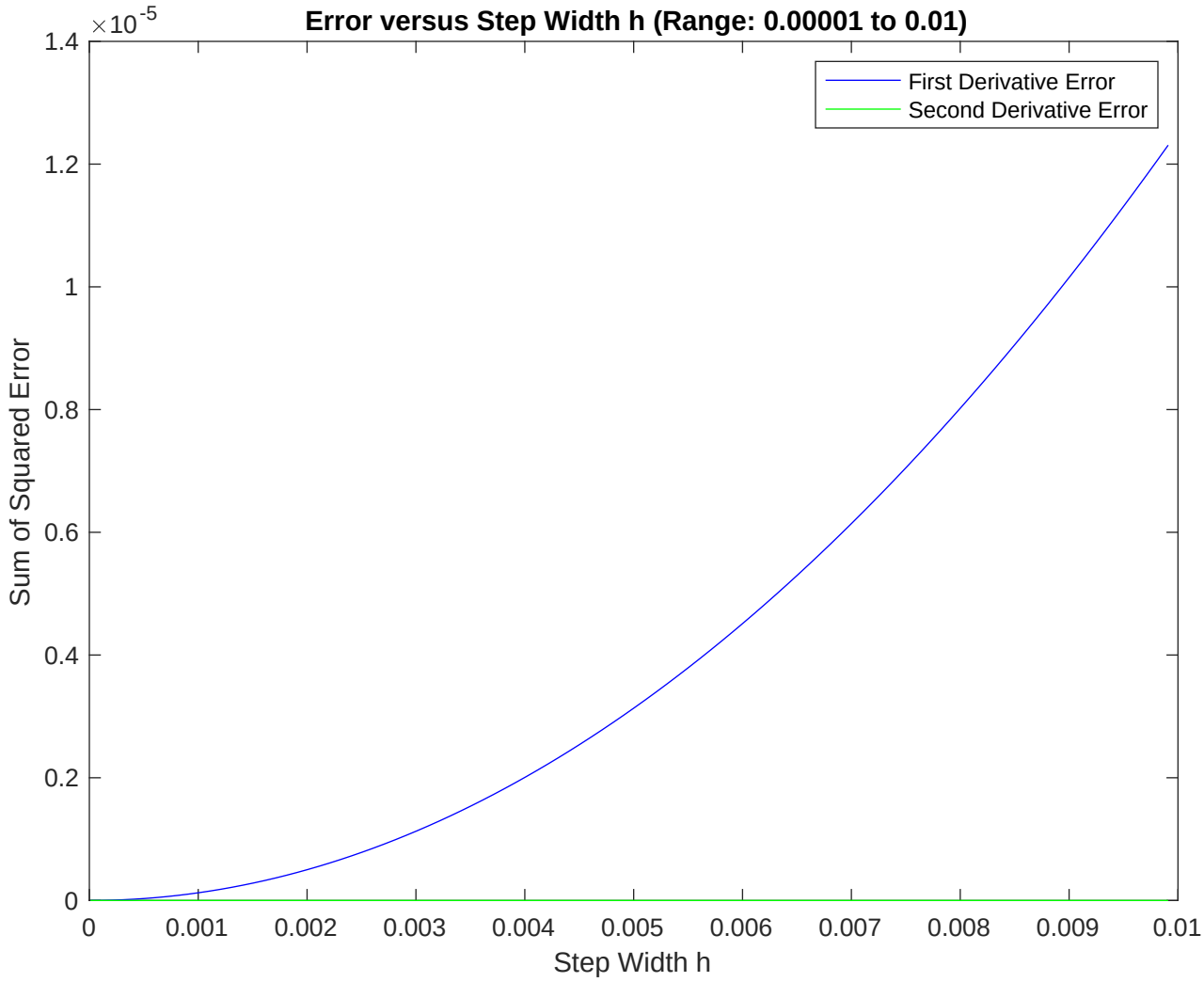
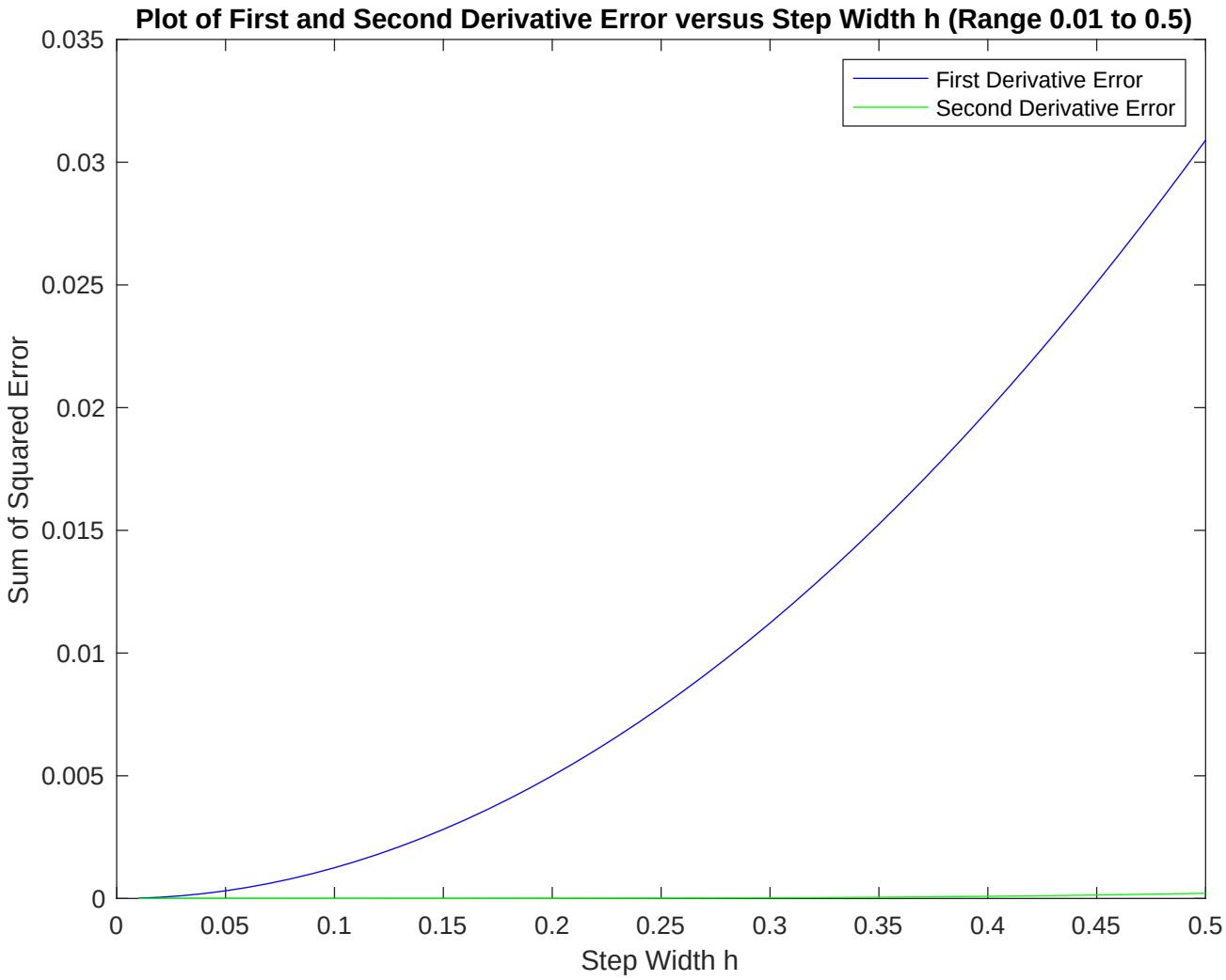

Plot of cos(x), Numerical derivative and -sin(x)
cos(x)
Numerical derivative
-sin(x)
y
x
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-4
-3
-2
-1
0
1
2
3
4


Plot of cos(x), Improved Numerical derivative and -sin(x)
cos(x)
Improved Numerical derivative
-sin(x)
y
x
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-4
-3
-2
-1
0
1
2
3
4


Plot of cos(x), Numerical second derivative and -cos(x)
cos(x)
Numerical second derivative
-cos(x)
y
x
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-4
-3
-2
-1
0
1
2
3
4


Error versus Step Width h (Range: 0.00001 to 0.01)
×10^-5
1.4
1.2
1
0.8
0.6
0.4
0.2
0
Sum of Squared Error
0
0.001
0.002
0.003
0.004
0.005
0.006
0.007
0.008
0.009
0.01
Step Width h
First Derivative Error
Second Derivative Error


Plot of First and Second Derivative Error versus Step Width h (Range 0.01 to 0.5)
First Derivative Error
Second Derivative Error
Sum of Squared Error
Step Width h
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5